

S P E C T R A V I D E O

U S E R S G R O U P -- Wgtn

Newsletter Number 21 -- May 1986

The next meeting is on Monday, May 19th at 7.00 in the Staff Common Room, Wellington Clinical School, Main Street Newtown, the next will be on 16 June 1986. (ie as usual the 3rd Monday of the month.)

Wellington SV Users Group

Committee Members

Chairman ----- Darryl Alison Ph 85522 (Paraparaumu)

Secretary ----- Joseph Dol 889275

Treasurer ----- Ross Fletcher 788454

Don Stanley 896379

Garry Clark 367872

Newsletter Committee

Steve Hanson 843495

John Matthews 327805

Other specific areas of responsibility for committee members are

Don -- Software Consultant

Garry -- Radio Access & public domain tape software;

John -- Library;

Darryl -- Program evaluation committee chairman & Kapiti users representative.

Palmerston North Subgroup

The Palmerston North subgroup are now holding meetings on the 1st Monday of each month at member's homes. For more information they suggest you contact:

Steve Lacy; C/o Kesta Electronics, WANGANUI. Ph (064) 55501(w) 47310(h).

Graneme & Stuart Merrill, 25 Heatley Ave., PALMERSTON NORTH. Ph (063) 71337.

Justine Pickering, 68 Wakeman Street PARIATUA. Ph (0650) 8390.

Graham Horne, 36 Miro Street, PALMERSTON NORTH. Ph (063) 87829.

GAMES SCORES

Spectron	12910	Robert Campin
Sector Alpha	2445200	Garry Clark
Kung Fu	71405	Garry Clark
	33575	Garry Clark (*)
S C Force	748600	Deirdre Chin
Telebunny	113520	Garry Clark
	32720	Garry Clark (+)
	31670	Kalima Collinson (*)
Omac	35360	Richard Bremford
	25700	Don Stanley (+)
Ninja	251350	Nicola Tuinman
Tetra	284450	Julian Stone
Turboat	68390	Garth Thornton (*)
Turboat	104610	Taane Clark
Sasa	172950	Steve Lacey
Frantic Fred	21720	Rachel Hughes (level4)
Flipper Slip	37640	Garth Thornton
Bats Alive	3540	Colin Chin (*)
Galaxy	1447	"
Jumping Jack	2600	"
TV Nightmare	7040	"
	3640	" (*)
	4980	Deirdre Chin (+)

(+) = started in normal level

(*) - started in arcade level

otherwise practice level or unknown.

Although it is nice not to have to update the scores all the time, it would be nice to see a few more changes in the scores above. C'MON all you gamesmasters - show us how good your skills are !!

Club Sponsor

Action Computers -- P.O Box 7442, Wellington South.

Photocopying from Library

Anybody requiring articles from the library to be photocopied please send request to the LIBRARIAN, not any other committee member. We have had problems with articles not being copied when requested, please ensure that requests go to the librarian.

Also membership renewals should be sent to Box 26050, Newlands, Wellington, not to any other box, otherwise it is possible that the mailing list will not get updated.

SV 32A Owners:

The Magic of SpectraVideo is still available for \$55.00. This includes an assembler on tape. Very limited numbers available so see Don - or your local dealer.

AND NOW FROM THE "YOU BLEW IT STEVE !!" DEPARTMENT

On page 10 of our last months newsletter, I included a **BUG** report on our Jan/Feb issue (The program to convert bytes into DATA statements) - I've since discovered an easier solution to the 'Out of String space' message - The program was using BASIC'S default String allocation of 200 bytes. To get more space, just add a 'CLEAR n' statement in the opening lines of the program, to clear sufficient string space - I leave it to you to work out how much.

-----+----- AND FROM THE "YOU BLEW IT EDITORS !!" DEPARTMENT

We must apologise to the majority of you who have only received your April Newsletter with this issue. We have lost our original newsletter printer, and are now having to farm the printing out to whoever will do it (at a reasonable cost). To have an issue printed by any outside organisation has either one or both of the following faults:

EITHER they charge twice what the club was charged previously.
OR they cannot give a swift turnaround of the job - thus delaying the Newsletter further.

As discussed at our April meeting we are making best efforts to overcome these problems but PLEASE bear with us in the meantime. It is club policy to ensure that our Monthly meeting is on the 3rd Monday of every month - so you shouldn't need the newsletter as a reminder of the meeting date - and we ALWAYS have at least a few copies of the newsletter at the meeting even if we have been unable to post them in time.

Club Subscriptions

The club subscription is now \$15.00/year

New members this month include Chris Fage, John McCosh, Ron Lynch, H. Eaton.

Current Membership 148

Note that expiry date for subscriptions is printed on the address label of newsletter. Subscriptions not renewed a month after this date will not receive newsletters from then.

The next couple of pages are part of a large contribution from Bob Curwood of Christchurch, which we will feature in the next few issues. Most of these require ASSEMBLING a Machine Code routine - using 'ZEN' or the 'Magic' assembler mentioned above:-

```

; New commands

; This program is a demo of how to
; add new commands to BASIC by
; prefacing them with the CMD
; command. In this example the
; following 4 commands are checked
; with a jump to 4 subroutines as
; appropriate. If there are any
; errors then the routine exits to
; the SYNTAX ERROR in XXX routine

```

```

;CMD TIME SET (or CMDTIMESSET)
;CMD TIME ON (or as above )
;CMD TIME OFF (" " " ")
;CMD TIME PRINT (CMD TIME?)

```

```

;Note the following points
;1. If the command is accepted the
;HL pair should be left pointing
;to the next byte after the CMD
;(whatever). Thus after the CMD &
;the TIME XXXX is accepted the HL
;pair is pushed, saving the
;address of the NEXT command on
;the stack & after the required
;subroutine is performed it is POP
;-ed and the return address to the
;CMD routine is removed by a dummy
;POP AF.

```

```

;2.The easiest way to implement
;a new command is to use one of
;the existing reserved words as
;they are tokenised & thus only
;1 or 2 bytes need checking.
;However, it is possible to make a
;completely new command word but
;it means that each letter needs
;checking.(Unless some of them are
;a reserved word, see the TO in
;the middle of MONITOR)

```

```

;3.You could also use the MON word
;by using the hook at FF84H.
;For example a new command of
;MONITOR would call your routine
;(via the hook) with the HL pair
;pointing to the start of ITOR.
;See the last example

```

```

;4.To use the CMD routine you must
;place a jump to your routine at
;the CMD hook, FF9CH.

```

```

;5. After the call to CMD (or MON)
;HL is pointing to the next non-
;space letter of your command,thus
;if you only have a 2 word command
;e.g CMD TIME then it doesn't
;matter how many spaces in between

```

```
;them but if you have 3 (or more!)
;then it is your job to check for
;a space & jump over if nessesary.
;See SPACE1
```

```
;6.The print routine is only used
; to demonstrate the result of a
; command.
```

```
;7. The RST 10H checks that the
; byte after the new command is
; either a ':' separating it from
; the next command or the end of
; the line, in either case it
; will return with the Z flag set
; if OK.
```

```
;To use the example TIME .ZEN first
;ASSEMBLE it and then use the M
;command (if you are using ZEN) to
;change the CMD hook to C3 1D 90 &
;change the MON hook to C3 75 90.
```

END

```
*****
;****
;****      CMD TIME SET, DISPLAY, ON, OFF      ****
;****
;****      by      ****
;****      Bob Curwood      ****
;****      Technical Officer      ****
;****      Christchurch Spectravideo      ****
;****      Users Group      ****
;****
;*****
```

```
ORG 9000H
LOAD 9000H
```

```
CLS: EQU 03777H
CHPUT: EQU 00018H
POSIT: EQU 0393EH
ERROR: EQU 00907H
CSRY: EQU 0FA03H
CMDHK: EQU 0FF9CH
```

```
TIME: EQU 0EFH
SET: EQU 0D2H
ON: EQU 095H
OFF: EQU 0EBH
PRINT: EQU 091H
```

```
TEST:      CALL ALTERHK
           JP 0A003H
```

;Jump to ZEN for now

```
;Delete above 2 lines after de-bugging
```

```
ALTERHK:   LD HL,CMDHK
           LD A,OC3H
           LD DE,CMD
           LD (HL),A
           INC HL
           LD (HL),E
```

```

INC HL
LD (HL),D
RET                                ;Return to BASIC

CMD:      LD A,(HL)                ;Jump here from CMD
          CP TIME
          JR NZ,EXIT3              ;Error exit

SPACE:    INC HL
          LD A,(HL)
          CP " "
          JR Z,SPACE               ;Jump over spaces
          PUSH HL                  ;Address of next command
          CP SET
          JR Z,SETIME
          CP ON
          JR Z,TIMEON
          CP OFF
          JR Z,TIMOFF
          CP PRINT
          JR Z,PRTIME
          JR EXIT3                  ;None of these!

SETIME:   LD DE,MS1
          LD HL,(CSRY)
          CALL PRINT1
          CALL POSIT
          JR EXIT4                  ;OK Exit

TIMEON:   LD DE,MS2
          LD HL,(CSRY)
          CALL PRINT1
          CALL POSIT
          JR EXIT4

TIMOFF:   LD DE,MS3
          LD HL,(CSRY)
          CALL PRINT1
          CALL POSIT
          JR EXIT4

PRTIME:   LD DE,MS4
          LD HL,(CSRY)
          CALL PRINT1
          CALL POSIT
          JR EXIT4

EXIT3:    LD E,2                    ;Syntax error
          JP ERROR

EXIT4:    POP HL
          RST 10H                  ;Check for : or 00 byte
          JR NZ,EXIT3
          POP AF
          RET

PRINT1:   PUSH HL
          EX DE,HL
          LD D,(HL)
          INC HL
          LD E,(HL)
          LD A,D

```

```

OR E
JR Z,LOOP2
EX DE,HL
CALL POSIT
EX DE,HL
LOOP2: INC HL
LD A,(HL)
AND A
JR Z,EXIT2
CALL CHPUT
JP LOOP2
EXIT2: POP HL
RET

MS1: DB 32,1,"SET TIME",0
MS2: DB 32,1,"TIME ON ",0
MS3: DB 32,1,"TIME OFF",0
MS4: DB 32,1,"12:45:18",0
END

```

```

;*****
;*****
;*****          New Error Messages          *****
;*****          by                               *****
;*****          Bob Curwood                     *****
;*****          of                               *****
;*****          Christchurch SpectraVideo       *****
;*****          Users Group                     *****
;*****          *****
;*****

```

```

;          SV328 Disk or Tape
          ORG 9000H
          LOAD 9000H

          ERRP: EQU OFF18H
          BACK: EQU 0EB13H      ;Where the hook used to go
          WRMZEN: EQU 0A003H    ; for disk bask errors.

TEST:      CALL ALTERHK
          JP WRMZEN            ;Delete these 2 when debugged

ALTERHK:   LD HL,ERRP
          LD A,0C3H            ;Jumpr to new routine
          LD DE,NEWERR

; , or OLDERR
          LD (HL),A
          INC HL
          LD (HL),E
          INC HL
          LD (HL),D
          RET                  ;Return to BASIC

NEWERR:    LD A,E
          CP 80                ;Lower limit of messages
          JR C,EXIT
          CP 90                ;Upper limit plus one
          JR NC,EXIT           ;in this case 10 new ones.
          SUB 79
          LD E,A
          LD HL,MSGBSE         ; Put into E reg
          RET                  ;New messages start here
          ;To routine that called ERRP

```

```

EXIT:          JP BACK          ;Check for disk error mesgs:
;If using tape-based OS then above exit is a RET
;OR change the 2 JR [NJC, EXITs to RET C (or NC) and delete the
; above line.

```

```

MSGBASE:       DB 0             ;Must have a zero here
               DB "Error message # 80",0
               DB "Error message # 81",0
               DB "Error message # 82",0
               DB "Error message # 83",0
               DB "Error message # 84",0
               DB "Error message # 85",0
               DB "Error message # 86",0
               DB "Error message # 87",0
               DB "Error message # 88",0
               DB "Error message # 89",0

```

```

OLDERR:        LD A,E           ;Use this routine instead to allow
               CP 2             ; any of the normal error messages
               JR Z,NEW2        ; to new ones, in this case I am
               CP 5             ; changing the 2 most often en-
               JR Z,NEW5        ; countered, SYNTAX ERROR and
               JR EXIT1         ; Illegal function call as a demo.
NEW2:          LD E,1           ; "First new message
               JR NEWLOC
NEWS:          LD E,2           ; Second new message
NEWLOC:        LD HL,NEWMESG    ; Location of new messages
EXIT1:         JP BACK         ; Back to the normal routine for
                               ; disk or simply RET for a tape sys:

```

```

NEWMESG:       DB 0
               DB "You've spelt it wrong, dummy!",0
               DB "Hey, I can't do that!",0

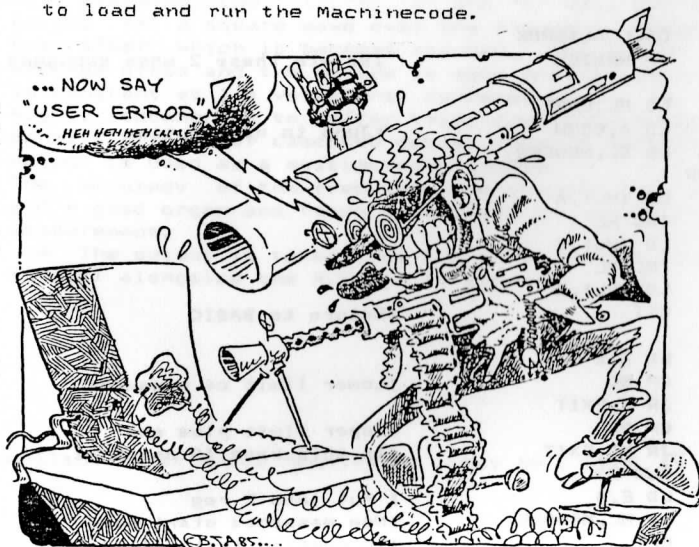
```

END

Editors Note:-

I have successfully made up a few machine code routines using pen, paper and listings of assembler mnemonics similar to those in the back of our December 1985 issue, then incorporating them into a program as DATA statements with a suitable loading routine to load and run the Machinecode.

Steve H.




```

10 WIDTH80:CLS
20 PRINT"This program sets up the function keys"
30 PRINT"to use the SV328 version of ZEN
40 PRINT
50 PRINT"ZEN is being loaded at &HA000
60 PRINT"from drive 1":PRINT
70 PRINT"Please wait
80 CLEAR 300,&H9FFF
90 BLOAD"1:ZEN328":POKE&H80C,74:POKE&HA105,&HC7:POKE&HA106,&HD2
100 KEY 1,"P22"+CHR$(13)
110 KEY 2,"T"+CHR$(13)+"P22"+CHR$(13)
120 KEY 3,"G9000H"+CHR$(13)
130 KEY 4,SPACE$(13)
140 KEY 5,"": EQU "
150 KEY 6,"X=USR(0)"+CHR$(13)
160 KEY 7,"GOTO310"+CHR$(13)
170 KEY 8,"DEFUSR=&HA000:"
180 KEY 9,"GOTO320"+CHR$(13)
190 KEY10,"GOTO350"+CHR$(13)
200 CLS
210 PRINT:PRINT"Use F6 to run ZEN"
220 PRINT:PRINT"Use F8 to DEFUSR if required
230 PRINT:PRINT"Use F7 key to save a ZEN text file (from
m BASIC)
240 PRINT:PRINT"NOTE A reset of the computer will (if usin
g a disk based system) destroy the contents of C100 to
C17F inclusive which is part of the ZEN text-file (if one exi
sts) MORAL! SAVE it before testing it!
250 PRINT"To reload it to ZEN use F9 key F10 will give
a printout of the source file (Make sure the last line of code
is 'END')" BY ITSELF, @ LEFT-MOST COLUMN
260 DEFUSR=&HA000
270 COLOR4,7:PLAY"o7BBBBDc":END
280 SAVE"1:ZEN":END
290 BSAVE"1:ZEN328",&HA000,&HBC00
300 END' -----
310 A$="1: " :B$="&HC000":C$="&H"+HEX$(PEEK(&HA10C))+RIGH
T$( "0"+HEX$(PEEK(&HA10B)),2):PRINT"BSAVE"+CHR$(34)A$CHR$(34)", "B
$,"C$STRING$(24,29):LINEINPUTD$:D$=D$+STRING$(ABS(9-LEN(D$)),3
2):D$=LEFT$(D$,9):BSAVE"1:"+D$,VAL(B$),VAL(C$):END' -----
320 PRINT"Enter file name>";LINEINPUTA$
330 BLOAD"1:"+A$
340 POKE&HA10B,PEEK(&HF9B4):POKE&HA10C,PEEK(&HF9B5):Z=USR(0):END
350 A=&HC000:IF (INP(18)AND1)=1THENPRINT"PRINTER not switched on/
ready":END
360 A$="":C=0
370 B$=CHR$(PEEK(A+C))
380 A$=A$+B$:C=C+1
390 IFB$<>CHR$(13)THEN370
400 FOR D=1TO1
410 LPRINTA$;CHR$(20);
420 NEXT
430 LPRINT
440 A=A+LEN(A$)
450 IFA$="END"+CHR$(13)THENENDELSE360
460 INPUT"TITLE>";Z$:GOSUB520:Z$=" ":GOSUB520
470 A=&H9000:C=&H9100'start/end
480 Y$="":Z$=HEX$(A)+" ":FOR B=0TO15:X$=RIGHT$("0"+HEX$(PEEK(A+
B)),2):Z$=Z$+X$+" "
490 IF VAL(X$)<32ORVAL(X$)>126THENX$="."ELSEX$=CHR$(VAL(X$))
500 Y$=Y$+X$
510 IFA+B>C-1THENGOSUB520:ENDELSENEXTB:GOSUB520:GOTO480
520 FORD=1TO3:LPRINTZ$;" "Y$CHR$(20);NEXT:LPRINT:A=A+16:RETURN
530 'Program by Bob Curwood. Christchurch NZ

```

Notes from Don Stanley

Magazine Review -- MSX & SpectraVideo Computer Forum

This magazine is available for MSX & SpectraVideo 318/328 owners. The first issue has just been released, and retails in NZ for \$4.00. PandaSoft International evidently are bringing it into NZ.

The magazine is published in Australia. It looks great for beginners and those who lack a little confidence in operating their computers. Also MSX owners will appreciate it. From an experienced 328 owners point of view, issue 1 didn't say anything new, indeed much of issue 1 has been presented in WGTN users groups magazines over the past 2 years.

It is fairly full of MSX news and developments. Of extreme use to the beginner is a liftout Basic tutorial covering both MSX and SVI. It is absolute beginner stuff, and good at it too, except for one glaring inconsistency ...

'.. What is wrong with the following program ?

```
100 print "Line 100"  
110 rem goto 140  
120 print "Line 120"  
130 print "Line 130" ....'
```

The answer is absolutely nothing is wrong, it would be if the REM wasn't there. This is the sort of question guaranteed to trip up the beginner. But this tutorial liftout looks good, and the self answer questions are good -- hope they issue solutions in the next issue though.

There are a few programs included, and also book reviews and advertisements. As I said it is currently beginner oriented, but will get more technical as advanced users submit articles. Subscription price is \$32.00 Australian, you can send subscriptions to Subscription Dept, P.O Box 71, Palmyra, W.A 6157 Australia. Worth investing in, provided computer owners support it with articles. I know from experience that that is something the average computer owner is far from good at!

Bits & Bytes had dropped the sprite generator program off the previous March article, and I had jazzed it up and sent it again for the June issue. Then they printed it in the MAY issue. Heres the program jazzed up a bit to tell you where the sprite characters are in memory. It may appear in June B & B also.

```
10 SCREEN 1  
20 FOR I=&H4198 TO &H4550 STEP 8  
25 LOCATE 10,10:PRINT HEX$(I)" "HEX$(I+7)  
30 FOR J=I TO I+7
```

```

40   A$=A$+CHR$(PEEK(J))
50   NEXT
60   SPRITE$(1) = A$
70   PUT SPRITE 1,(128,96),7,1
80   A$=INPUT$(1) : A$="":CLS
90   NEXT

```

The SVI character set is from 4198h to 4550h in rom. Each character takes 8 bytes, each byte corresponding to one of the 8 rows required to build the full character. For MSX change the 4198 to 1BBF, the 4550 to 23B7, and the SCREEN 1 to SCREEN 2. The program will show every character on screen as a sprite, 1 at a time and where in memory the character resides. MSX users will need to open the GRP: for output and write to that to get the locations output.

Software Reviews -- MSX

Jetset Willy -- Superb game, Willy has to collect all the flashing cups and things from 20 odd rooms by midnight before the maid will let him go to bed after his wild party. This game is really addictive, really difficult, and great fun. Every room has something horrible trying to stop you collecting the goodies and some of them are in impossible places. Costs \$27.95 on tape. Xpress owners Action Computers can supply on 3.5" disk for \$11.00, you must include proof of purchase on tape. Proof means either a receipt including both your name and the program name, or some other proof from your dealer!

Manic Miner -- very similar to Jetset Willy except you have to clear each room before going to a new one. Same ideas, some of the same monsters, and of course, written by the same company. Both of these programs were original Spectrum best sellers, and will appeal to many MSX owners. Price same as Jetset Willy.

SuperChess -- a very good version of chess which has hammered everyone I've let loose on it so far. Plays moves in 3 seconds at normal level, or up to 6 hours at competition level. Retail price is \$24.95 (at CDL's last price), again I can supply on 3.5" disk for \$11.00 and same conditions as above.

River Raid -- an excellent shoot em up as you fly down the river type game. Evidently there are 50 levels, you have to avoid bridges, steep terrain, and nasties shooting as you as you head for the other end. Last CDL price was \$54.95 which I thought a bit excessive.

Hobbit -- to my disgust and anguish this just won't load on the Xpress. It loads and runs perfectly on the 728 though, and I'm

told its a superb game. It should be at \$69.95 including the paperback.

Tennis -- one of the overpriced MSX cartridges, but the more I play it the more my feelings about the high price crumble. This is excellent, and evidently very popular. Superb graphics and sound and the coolest ball boy around make this game thoroughly addictive. \$85.00 at last CDL price.

I was recently asked about loading tapes into CPM. Action Computers program BEEMON allows you to do just that. Tapes of any type (BASIC, MCODE, SCREENDUMP, ASCII) can be loaded and then the contents either saved to CPM disk or edited. Also Color Genie/TRS80 and MicroBee tapes can be used. Many other features such as complete disassembly, memory search, trace and so on are offered also.

The Canadian Users Group have very kindly sent a disk full of MSX public domain programs. But some are in French and incomprehensible to people who speak standard computer languages. So if anyone wants these (on 3.5" disk only) be prepared to wile some of the hours with a translation dictionary. Some look quite good.

Zork2, Zork3 and Hitch Hikers Guide to the Galaxy can be obtained for the 328 under CPM. I have the Zorks and they are so difficult! Anyone know how to move a menhir, without help from Obelisk? By the way, once again let me stress that if you buy CPM software make sure it is in XEROX 820-2 ss/dd or XEROX 820-2 ds/dd format. If you can't get that, go for Kaypro 2 format and I can transfer the software to Xerox 820-2 for you.

Back next month , Don Stanley



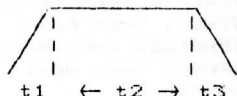
SOME SOUND THOUGHTS

By Jim Oliver

The SV-328 claims to have an Attack, Decay, Sustain, Release (ADSR) sound envelope. Possibly this is debatable but in any event it is possible to obtain good control of these features by using the SOUND registers 8, 9 and 10. These are the 'Volume Controllers' for the 3 Sound channels. This is demonstrated by the next program.

```
10 A=0:B=11:C=0.1
20 FOR V= A TO B STEP C
30 SOUND 7, 248
40 SOUND 8, V: SOUND 9,V: SOUND 10, V
50 SOUND 1, 1: SOUND 0, 220
60 SOUND 3, 1: SOUND 2, 123
70 SOUND 5, 1: SOUND 4, 62
80 NEXT
90 FOR T = 1 TO 900: NEXT T: IF V < 1 THEN END
100 A = 11: B = 0: C = -C
110 GOTO 20
```

This plays the notes "C", "G" and "E" in 3 separate channels at the same time. The envelope shape is something like this:



T1 is controlled by the initial value of C (.1). T2 is controlled by the value of T in line 90. T3 is controlled by the value assigned to C in line 100. Various plateaux may be inserted by adding more for-next loops in the correct places and testing for the values of V along the way.

In theory I could have made the initial value of B = 15. If you try it you will see why I didn't.

Anyone who has experimented with PLAY or SOUND will know the distortion produced by this:

```
PLAY "SOM5000C","SOM5000G","SOM5000E"
( Compare M=2000 and M=3000 )
```

I don't know how this distortion occurs or if there is any way (other than the above) of preventing it. I would be interested to hear from anyone who knows more.

While we are on the subject try these:

```
10 A%=INPUT$(1)
20 A=ASC(A%)/3
30 B=A+12:C=B+12
40 PLAY "T200L32","T200L32","T200L32"
50 PLAY "N=A;","N=B;","N=C;"
60 GOTO 10
```

This gives quite a convincing pipe organ and could be used as the basis of (say) "Spectravideo Organ"

Thousands (at least) of different sounds are possible with the SOUND command. Here's yet another:

```
10 A%=INPUT$(1)
20 A=ASC(A%)
30 B=A-100:C=B-60
40 FOR I=11 TO 0 STEP -.2
50 FOR D=5 TO 1 STEP -.1
60 SOUND 8,I:SOUND 9,I:SOUND 10,I
70 SOUND 7,56:SOUND 1,D:SOUND 0,A
80 SOUND 3,D:SOUND 2,B
90 SOUND 5,D:SOUND 4,C 100 NEXT D
110 NEXT I
120 GOTO 10
```

You should get a weird echoy sound after pressing any key! Try pressing a function key. There are endless possibilities.

Finally, for the person who wants a programmable square wave generator but can't be bothered building it, try this:

```
10 CLS: CLICK OFF: DEFMSG H
20 AC=0: AF=50
30 A%=INKEY$: IF A%="" THEN GOTO 30
40 IF A%="1" THEN AC=AC+1: IFAC>15 THEN AC=15
50 IF A%="h" THEN AC=AC-1: IFAC<1 THEN AC=0
60 IF A%="d" THEN AF=AF+1: IFAF>255 THEN AF=255
70 IF A%="i" THEN AF=AF-1: IFAF<1 THEN AF=1
75 HZ=3579545#/(16*((256*AC)+AF))
80 LOCATE 8,5: PRINT "Coarse";AC
90 LOCATE 20,5: PRINT "Fine";AF
100 LOCATE 8,7: PRINT "Freq in Hz =";HZ
110 SOUND 7,254: SOUND 8,9: SOUND 1,AC: SOUND 0,AF
120 GOTO 30
```

By pressing the "1","h","d" and "i" keys it is possible to output (*) a square wave over the frequency range 54.6 to 223.7 KHz after which it becomes rounded, and at 200 KHz the output voltage drops and the square is much more triangular! The voltage is constant at 1.4 volts(p-p) centered around 10.7 volts. It is highly recommended to buffer this "raw" output with a simple CMOS Schmitt trigger or cascaded gates. This is particularly so if the output is used as a multipurpose source. The accuracy of the frequency was checked with an oscilloscope and a good organ and found to be spot on, accepting the errors of measurement.

* The output is taken from the audio output (above the video output) alongside the R.F. or TV output.

CUSTOMISING WORDSTAR

John Bridson

Ross Fletcher's article on re-installing WordStar 3.30 and Graeme Dick's on customising WordStar's Print Menu were both published in the club magazine for Jan/Feb 1986. The experiences they reported are similar to mine, but there have been several more points which I have come across and/or have been asked about by others and which seem worth sharing.

In what follows, all codes are given in hexadecimal form.

INSTALLING WORDSTAR 3.30

The sections below list the values used in my version of WordStar for each of the possibilities available when using WINSTALL. To check your version, start up WINSTALL as per the manual and follow the screen prompts to get to the *Installation Menu*. Don't forget to tell WINSTALL if some of your files (WS.COM, for example) are not on the logged (default) drive, and DON'T use WS.COM for the name of your new version of WordStar, in case it doesn't work properly! Use a name like WSTEST.COM instead and rename it later, when you are convinced it's OK. (For SpellStar to run properly, this has to be done using WINSTALL to reinstall WSTEST.COM as WS.COM).

Installing the Terminal

For WordStar to interface correctly with the computer's terminal (display driver) functions, the program needs to know what codes are required for certain screen operations. The type of monitor used has no relevance to this process: for the record, mine is a green screen Zenith ZVM-123-E.

The selection to make at the Installation Menu is **B, Custom Installation of Terminals**: the *Terminal Installation Menu* uses the headings below.

B Terminal Name (optional)	Spectravideo SVI-328 (preceded by 7 spaces, to centralise it)
C Screen Size	Lines? 24 Columns? 80
D Cursor Positioning	Code for cursor about to move? 1B 59 Code sent between move coordinates? none (empty) Code sent following a move? none (empty) Code for line 1? 20 Code for column 1? 20 Code type used for line, column numbers? binary Order line, column numbers sent? line first
E Terminal Start-up	Code needed when starting WordStar? none (empty)
F Terminal Exit	Code required when stopping WordStar? 0C ** ** Causes a Clear Screen when leaving WordStar

G Highlighting	Code to turn highlight ON? 1B 70 Code to turn highlight OFF? 1B 71
H Erase to End of Line	Code to erase all characters from cursor position to the end of the line? 1B 4B
I Delete Line	Code to erase present line and move all following lines up? 1B 4D
J Insert Line	Code to insert blank line at present cursor line and move all others down? 1B 4C
K Handling of Last Character on Screen	Screen scrolls up a line when cursor passes the bottom right screen corner? yes

Installing the Printer

For WordStar to interface properly with your printer and use as many of its features as possible, the program needs to know what codes must be sent to the printer for each of those features. My printer is a dot-matrix Logitec FT-5001 whose control codes are virtually completely Epson-compatible. The Logitec is almost identical to the Super 5 EP-1201 and Panasonic KX-P1091 printers, except that it has no NLQ capability.

The selection to make at the Installation Menu is *D, Custom Installation of Printers*: the *Printer Installation Menu* uses the headings below.

C Printer Name (optional)	Logitec FT-5001 (preceded by 9 spaces, to centralise it)
D Initialisation	Code required when starting print? 0D Code required when ending print? none (empty)
E Overprinting	Does printer back-space? choose 'C' (printer CAN backspace, but choosing the NON-BACKSPACING option is much easier on the printer!)
F Boldfacing	Number of passes for boldface print? 3
G Protocol Menu	Protocol required? choose 'A' (No protocol)
H Driver Menu	Printer Driver? choose 'A' (primary list device)
I Font (was Ribbon) Selection	Code to select alternate font? 1B 34 (ITALICS) Code to select standard font? 1B 35
J Vertical Motion	Code for platen about to move? none (empty) Code sent following a platen move? none (empty) Minimum VMI value? 1 Range of VMI values? 125 Code for line feed? 0A Code for reverse line feed? none (empty)
K Horizontal Motion	Code for print head about to move? none (empty) Print head horizontal increment? 1/120

- K Horizontal Motion Minimum HMI value? 1
 (continued) Range of HMI values? 125
 Code for space forward? 20
 Code for space backwards? 08
- L Print Modes Code for forward printing? none (empty)
 Code for backward printing? none (empty)
- M Phantom Characters Code for phantom space? none (empty) **
 Code for phantom rubout? none (empty) **
- ** any code entered here is printed as a space or backspace instead
- N Return/Line Feed Code for carriage return + line feed? 0D 0A
 Code for carriage return only? 0D
 Code for CR + 1/2 line feed? none (empty)
- O User-defined Code for ^PQ? 1B 2D 01 (Printer's UNDERLINE ON)
 Functions Code for ^PW? 0E ** (Wide Text ON)
 Code for ^PE? 1B 48 1B 54 (Sub/Superscript OFF)
 Code for ^PR? 1B 2D 00 (Printer's UNDERLINE OFF)
- ** Normal Text is automatically restored when a CR/LF occurs
- P Carriage Roll Code for roll up? 1B 53 00 (Superscript ON)
 Code for roll down? 1B 53 01 (Subscript ON)
- Q Character Pitch Code for alternate pitch? 0F (CONDENSED PRINT)
 Code for standard pitch? 12 (Standard pitch is
 ELITE style, 12 chars per inch)

Installing WordStar Features

The selection to make at the Installation Menu is *E, Menu of WordStar Features*: the *Features Menu* uses the headings below.

I've changed just a few of the features from the default settings, mainly to optimise printed output with my preference for the ELITE type style (12 characters per inch), rather than the wider (10 chars per inch) PICA style.

- J Bottom Margin Number of lines left blank at page foot? 6
- L Right Margin Column number of right margin position? 75
- N Page Offset Number of printer columns blank before Left Margin
 column is printed? 5
- R System Disk Drive Disk Drive Number to check for .OVR files, as well
 as logged drive? 1

Customising WordStar 3.30

If you want to customise your version of WordStar along the lines suggested by Deane Whitmore in his articles in the March and April 1986 magazines,

but are not too sure about using DDT, try using WINSTALL instead! The feature to use is the *Custom Modification Routine*, which allows you to change any of the first 1850 bytes of WS.COM (from address 0100h to address 0B3Ah). This is not documented in any recent WordStar manual (I came across it in a recent issue of PC Magazine), and you need to know either the hex address or the equivalent "label" name as well as the new data to insert before you can use it. Fortunately, Deane provided all of these items in his articles.

The selection to make at the Installation Menu is not shown: pressing the + key brings up the *Custom Modification Routine*. From here, just follow the instructions given on the screen.

Select the addresses you are interested in from Deane's articles and try different settings for each. All of these are listed below, with their original and my present values. Those label names in lower case can also be changed with WINSTALL's Features, Printer or Terminal setup functions. (The address given for USR4 is different from that Deane listed: the value is that WINSTALL V2.00 found when searching for the label name).

If you use the label name rather than the address, note that WINSTALL requires it to be preceded by a colon (:) during the data entry procedure.

Label	Address	Default	New Value	Label	Address	Default	New Value
bldstr	069A	02	03	initwf+01	036E	FF	FF
DBLSTR	069B	02	02	initwf+04	0371	FF	FF
DELCUS	028E	0A	01	INITWF+09	0376	01	01
DELMIS	028F	05	02	ithelp	034D	03	03
DEL1	02AF	03	01	itpopn	03FD	00	00
DEL2	02B0	09	03	PAGFIL	03EB	2D	2D
DEL3	02B1	19	06	SCRLSZ	02BA	14	2B
DEL4	02B2	40	0A	trmuni	027E	00	01 0C
DEL5	02B3	09	03	usr1	06CF	As ^PQ for printer	
initlm	036A	00	00	usr2	06D4	As ^PW for printer	
initpf+18	0369	08	05	usr3	06D9	As ^PE for printer	
initrm	036B	40	4A	usr4	06DE	As ^PR for printer	

WINSTALL can also be used to insert the "Printer-busy Test" Deane included in the April issue. Use EITHER the label names OR the addresses to tell WINSTALL where the patch is to go and enter the data. (MORPAT is actually at 02CBh, so the label call would be MORPAT+45). I haven't done this patch yet, but it should work just as well done this way. (Note that EH, given as data for address 0310, is not a valid hex value).

CUSTOMISING the WSMSSGS.OVR FILE for WORDSTAR 3.30

WordStar's Print Menu has been modified to match my printer commands. Some of my changes are similar to Graeme's, but I have also rearranged some of the other entries into a more logical order. Further, by removing some entries and combining others, I have added details about the User Patches.

As well as the Print Menu, I have also made minor amendments to the Help data, to reflect my choice of some of WordStar's Features.

The procedure I used to modify WSMSSGS.OVR is the same one that Graeme used. With DDT.COM and WSMSSGS.OVR on your disk, type

DDT WSMSSGS.OVR (DDT B:WSMSSGS.OVR if WSMSSGS.OVR is on drive B).

When DDT signs on, note the number (hex) shown under the heading NEXT (7680 in my case). You need to know this for saving your new WSMSSGS.OVR later.

Use DDT's S (Set) command to make those changes you want to include, using the form S<address>. To alter the code at a particular address, type the new value and then press <ENTER>. To leave it unchanged, pressing <ENTER> will bring up the next address. When there are no more to do, press <.> (period) to exit the S command.

Use DDT's D (Display) command, in the form D<address>, to check the changes you've made to the file. When satisfied that they're all correct, exit from DDT by typing ^C, ready to save your new version of WSMSSGS.OVR.

The CP/M SAVE command needs to know (in decimal) how many blocks of memory to move to disk. The two most significant digits shown under NEXT when DDT signed on and noted earlier (76 in my case) are the hex equivalent of the number required: this converts to 118 in decimal. Type

SAVE 118 WSMSSGS.OVR (SAVE 118 B:WSMSSGS.OVR to save it to drive B)

and you've finished, ready to try out your "New and Improved Version"!

The changes I made are given in address order below, using a format similar to that Graeme used.

Start Address 1B52

```
was  A   A l t e r n a t e   p i t c h
    41 20 41 6C 74 65 72 6E 61 74 65 20 70 69 74 63 68 20 20
now  A   B e g   2 0   C P I   E n d   N
    41 20 42 65 67 20 32 30 20 43 50 49 20 45 6E 64 20 4E 20
```

Start Address 1B8D

```
was  H   O v e r p r i n t   c h a r
    48 20 4F 76 65 72 70 72 69 6E 74 20 63 68 61 72 20 20
now  E   S t o p   S ' s c r i p t
    45 20 53 74 6F 70 20 53 27 73 63 72 69 70 74 20 20 20
```

Start Address 1BA2

was N S t a n d a r d p i t c h
4E 20 53 74 61 6E 64 61 72 64 20 70 69 74 63 68 20 20 20

now (S t d . P i t c h 1 2 C P I)
28 53 74 64 2E 20 50 69 74 63 68 20 31 32 20 43 50 49 29

Start Address 1BDD

was O N o n - b r e a k s p a c e
4F 20 4E 6F 6E 2D 62 72 65 61 6B 20 73 70 61 63 65 20

now H O v e r p r i n t c h a r
48 20 4F 76 65 72 70 72 69 6E 74 20 63 68 61 72 20 20

Start Address 1BF4

was P r i n t i n g P a u s e
50 72 69 6E 74 69 6E 67 20 70 61 75 73 65 20 20 20

now S t a r t P r i n t P a u s e
53 74 61 72 74 20 50 72 69 6E 74 20 50 61 75 73 65

Start Address 1C1C

was X S t r i k e o u t
58 20 53 74 72 69 6B 65 6F 75 74 20 20 20

now T S u p e r s c r i p t
54 20 53 75 70 65 72 73 63 72 69 70 74 20

Start Address 1C2D

was F P h a n t o m s p a c e
46 20 50 68 61 6E 74 6F 6D 20 73 70 61 63 65 20 20 20

now O N o n - b r e a k s p a c e
4F 20 4E 6F 6E 2D 62 72 65 61 6B 20 73 70 61 63 65 20

Start Address 1C42

was Y O t h e r r i b b o n c o l o r
59 20 4F 74 68 65 72 20 72 69 62 62 6F 6E 20 63 6F 6C 6F 72

now Q B e g P r n t e r U ' l i n e
51 20 42 65 67 20 50 72 6E 74 65 72 20 55 27 6C 69 6E 65 20

Start Address 1C7D

was G P h a n t o m r u b o u t
47 20 50 68 61 6E 74 6F 6D 20 72 75 62 6F 75 74 20 20

now W S t a r t W i d e P r i n t
57 20 53 74 61 72 74 20 57 69 64 65 20 50 72 69 6E 74

Start Address 1C92

was - - U s e r P a t c h e s - - **
A0 AD AD D5 F3 E5 F2 A0 A0 D0 E1 F4 E3 E8 E5 F3 AD AD A0 20

now R E n d P r n t e r U ' l i n e
52 20 45 6E 64 20 50 72 6E 74 65 72 20 55 27 6C 69 6E 65 20

** This line is in reverse video when the Menu appears on the screen, so the MSB obviously sets reverse video ON

Start Address 1CB0

was T S u p e r s c r i p t
54 20 53 75 70 65 72 73 63 72 69 70 74 20

now X S t r i k e o u t
58 20 53 74 72 69 6B 65 6F 75 74 20 20 20

Start Address 1CE2

was Q (1) W (2) E (3) R (4)
51 28 31 29 20 57 28 32 29 20 45 28 33 29 20 52 28 34 29 20

now Y A l t . F o n t B e g / E n d
59 20 41 6C 74 2E 20 46 6F 6E 74 20 42 65 67 2F 45 6E 64 20

Start Address 477A

was (d e f a u l t 8)
20 28 64 65 66 61 75 6C 74 20 38 29 20

now (d e f a u l t 5)
20 28 64 65 66 61 75 6C 74 20 35 29 20

Start Address 48F9

was e n d p a p e r (8)
65 6E 64 20 70 61 70 65 72 20 20 28 38 29 20

now e n d p a p e r (6)
65 6E 64 20 70 61 70 65 72 20 20 28 36 29 20

Start Address 4930

was d e f a u l t s t o 5 5)
64 65 66 61 75 6C 74 73 20 74 6F 20 35 35 29 20

now d e f a u l t s t o 5 7)
64 65 66 61 75 6C 74 73 20 74 6F 20 35 37 29 20